

EXPERIMENT-7b

Thivyaa P 240701570

Develop Low-Fidelity Paper Prototypes for a Banking App and Convert Them into Digital Wireframes Using Inkscape

AIM

The aim of this experiment is to construct low-fidelity paper prototypes for a banking application and digitize them into wireframes using Inkscape, a free and open-source vector graphics editor. The experiment covers the complete UI wireframing workflow from hand-drawn sketches to structured digital wireframes with proper shapes, labels, and navigation indicators.

Tools Used

- Inkscape – for creating digital vector wireframes
- Paper and Pencil – for initial low-fidelity sketches
- Tool Link: <https://inkscape.org/>

Procedure

Step 1: Create Low-Fidelity Paper Prototypes

1. Identify Core Features:

The banking application is designed to allow users to manage their accounts, perform transactions, and access banking services. The following core features and screens were identified:

- Register Screen – New account opening form with personal details and KYC fields.
- Home Dashboard – Net worth overview card, services grid, and promotional offers.
- Fixed Deposit Screen – View existing FDs and open a new fixed deposit.
- UPI Payment Screen – QR code display, UPI ID entry, and recent contacts.
- Notifications Screen – Filterable alerts for transactions, security, and offers.
- Profile / Edit Screen – Personal details editing with KYC status indicator.

2. Sketch Basic Layouts:

Plain paper and a pencil were used to sketch the basic layout of each screen. The focus was on placement of primary elements — buttons, input fields, navigation menus, and cards — without worrying about colours or visual styling. Rough boxes and labels represented all UI components.

- Simple rectangles were drawn for input fields, cards, and buttons.
- Navigation bars were sketched at the top and bottom of each screen.
- Labels and placeholder text were written inside elements to describe their function.
- Arrows were drawn between screens to indicate navigation flows.

3. Iterate and Refine:

The paper sketches were reviewed with peers to gather feedback before converting to digital wireframes. The following improvements were made based on feedback:

- Register Screen: Added terms and conditions checkbox and trust badge row below the form.
- Home Dashboard: Net Worth Overview card added at the top to show savings, FD, and total balance at a glance.
- Fixed Deposit Screen: Existing FDs section added above the new FD form so users can review before opening another.
- Notifications: Colour-coded left border added to each notification card for quick category identification.

Step 2: Convert Paper Prototypes to Digital Wireframes Using Inkscape

1. Install Inkscape:

Inkscape was downloaded from the official website (<https://inkscape.org/>) and installed. A blank document was created to begin wireframing.

2. Create a New Document:

A new document was created in Inkscape using File > New. A mobile canvas size of 360 × 640 pixels was set to match standard smartphone screen dimensions.

3. Set Up the Document:

- Document dimensions were set via File > Document Properties (360 × 640 px).
- A grid was enabled via View > Page Grid to assist with element alignment.
- Snapping was turned on to ensure elements aligned to grid lines consistently.

4. Draw Basic Shapes:

Inkscape's rectangle and ellipse tools were used to draw all UI elements. The wireframe style used thin black outlines with light grey fills to represent the interface in a neutral, uncoloured format:

- Rectangle tool (R): Used for input fields, buttons, cards, navigation bars, and containers.
- Ellipse tool (E): Used for avatar placeholders, icon circles, and toggle buttons.
- Line tool: Used for dividers, underlines, and navigation flow arrows.
- Bezier/Pen tool: Used for custom icons and arrow connectors between screens.

5. Add Text:

The text tool (T) was used to add labels, headings, placeholder text, and annotations to every element. Font size 10–14px was used for labels; bold 16px for screen titles and section headings.

6. Organise and Align:

- Elements were grouped by screen section using Object > Group (Ctrl+G).
- The Align and Distribute panel (Shift+Ctrl+A) was used to align elements horizontally and vertically.
- Equal spacing was applied between repeated elements such as form fields and list items.

7. Group Elements:

Related UI components on each screen were grouped together to keep the canvas organised. For example, the navigation bar, the balance card, and the transaction list were each grouped separately. This made it easy to move and edit each section independently.

8. Create Multiple Screens:

Six screens were created for the PurpleBank Banking Application by duplicating the base layout using Edit > Duplicate (Ctrl+D) and modifying the content for each screen:

- Screen 1: Register Screen – New account opening with personal and KYC details.
- Screen 2: Home Dashboard Screen – Net worth overview, services grid, and offers.
- Screen 3: Fixed Deposit Screen – View existing FDs and open a new FD.
- Screen 4: UPI Payment Screen – QR code, UPI ID entry, and recent contacts.
- Screen 5: Notifications Screen – Filtered alerts for transactions, security, and offers.
- Screen 6: Profile / Edit Screen – Editable personal details and KYC status.

9. Link Screens (Optional):

Navigation flow arrows were added between screens using Inkscape's Bezier tool. Arrows were drawn with labels to indicate user actions that trigger navigation:

- Create Account button → Home Dashboard
- Already have an account? Login → Login Screen
- FD/RD service icon on Dashboard → Fixed Deposit Screen
- UPI Pay icon on Dashboard → UPI Payment Screen
- Bell icon in navigation → Notifications Screen
- Profile tab in navigation → Edit Profile Screen
- Back button on all sub-screens → Home Dashboard

10. Export Wireframes:

Each screen was exported as a PNG image using File > Export PNG Image (Shift+Ctrl+E). Each page was selected individually, the export area was set to "Page", and a 96 dpi resolution was applied. The exported files were saved for documentation.

Digital Wireframes – Output Screens

The following screenshots show the six digital wireframes created in Inkscape for the banking application:

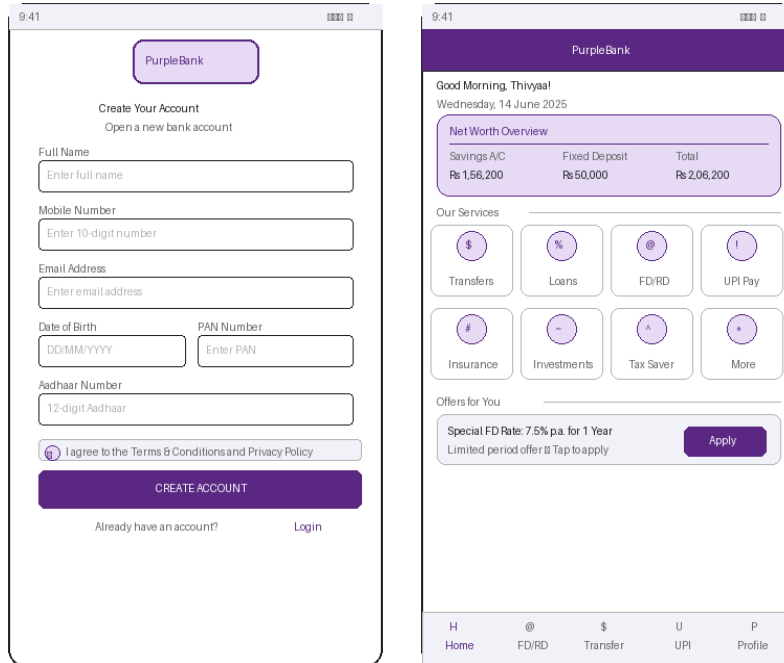


Fig 1: Register Screen (left) | Home Dashboard Screen (right)

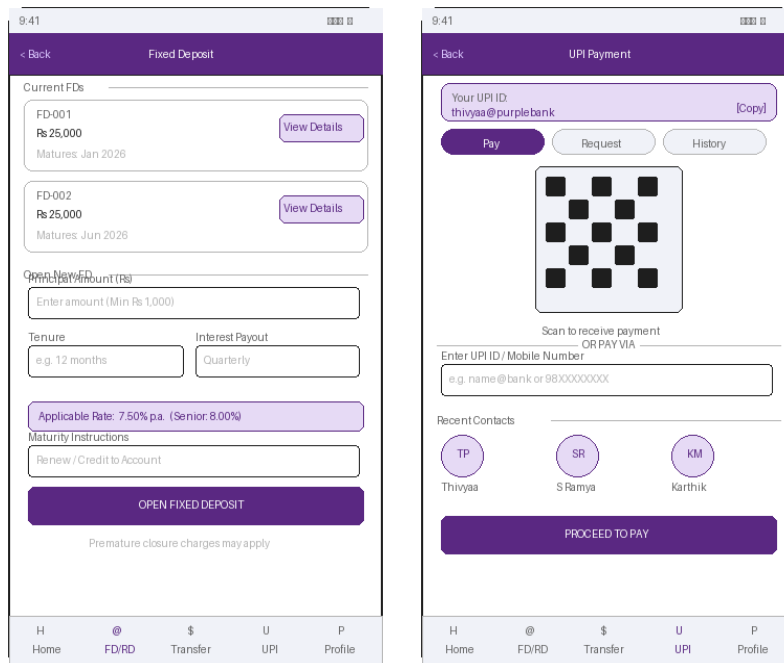


Fig 2: Fixed Deposit Screen (left) | UPI Payment Screen (right)

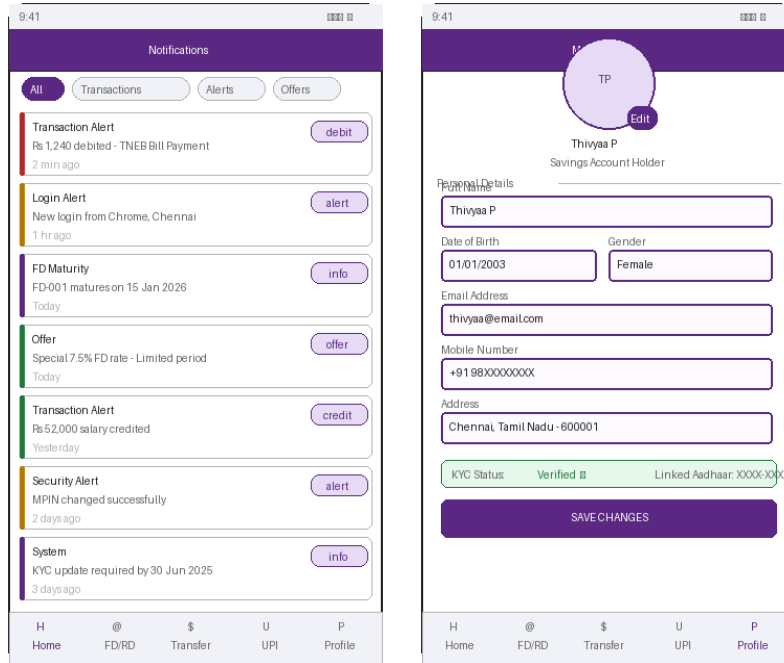


Fig 3: Notifications Screen (left) | Profile / Edit Screen (right)

Screen Descriptions

Screen 1: Register Screen

The Register screen enables new users to open a bank account within the app. The PurpleBank logo is displayed at the top for branding. The form includes fields for Full Name, Mobile Number, Email Address, Date of Birth, PAN Number, and Aadhaar Number. Date of Birth and PAN fields are placed side-by-side to reduce vertical space. A terms and conditions checkbox is included before the "Create Account" button. A login link at the bottom directs existing users back to the Login screen. The form uses a clean, minimal layout to avoid overwhelming first-time users.

Screen 2: Home Dashboard Screen

The Home Dashboard provides a personalised financial overview. A greeting and date are shown at the top. The Net Worth Overview card displays balances across Savings Account, Fixed Deposit, and Total in a three-column layout. Below it, an 8-icon Services grid gives quick access to Transfers, Loans, FD/RD, UPI Pay, Insurance, Investments, Tax Saver, and More. An Offers for You banner at the bottom highlights current promotional offers such as special FD interest rates with an Apply button. The bottom navigation bar provides persistent access to Home, FD/RD, Transfer, UPI, and Profile.

Screen 3: Fixed Deposit Screen

The Fixed Deposit screen is divided into two sections. The top section lists the user's existing FDs — FD-001 and FD-002 — each showing the principal amount, interest rate per annum, and maturity date, with a "View Details" button. The bottom section provides a form to open a new Fixed Deposit, with fields for Principal Amount, Tenure, Interest Payout frequency, and Maturity Instructions. An applicable rate banner dynamically shows the current rate (7.50% p.a., 8.00% for senior citizens). The "Open Fixed Deposit" button submits the form, with a note about premature closure charges.

Screen 4: UPI Payment Screen

The UPI Payment screen shows the user's UPI ID at the top with a Copy button. Three tabs — Pay, Request, and History — allow switching between UPI actions. A QR code placeholder is displayed centrally for receiving payments. Below a divider, a text field allows users to enter a recipient's UPI ID or mobile number for direct payment. A Recent Contacts row shows avatar circles with initials and names for quick repeat payments. The "Proceed to Pay" button at the bottom initiates the payment flow. The purple-accented design is consistent with the PurpleBank theme.

Screen 5: Notifications Screen

The Notifications screen presents all alerts in a filterable list. Pill-shaped filter chips at the top allow users to view All notifications or filter by Transactions, Alerts, or Offers. Each notification card displays a title, message, timestamp, and a category tag badge in the top-right corner. A colour-coded left border on each card provides instant visual categorisation: red for debits, orange for alerts, green for credits, and purple for informational notices. The list is ordered chronologically with the most recent notifications at the top.

Screen 6: Profile / Edit Screen

The Profile screen displays the user's personal information in an editable form layout. An avatar placeholder at the top shows the user's initials (TP) with an Edit button to change the profile photo. Below the avatar, the user's name and account type are displayed. The Personal Details section shows pre-filled editable fields for Full Name, Date of Birth, Gender, Email Address, Mobile Number, and Address. All fields are shown in their filled/active state with purple borders. A KYC Status bar at the bottom shows "Verified ✓" in green with the linked Aadhaar details. The "Save Changes" button submits any updates.

Result

The experiment was completed successfully. Low-fidelity paper prototypes were created for six screens of the PurpleBank Banking Application: Register Screen, Home Dashboard Screen, Fixed Deposit Screen, UPI Payment Screen, Notifications Screen, Profile / Edit Screen. These paper sketches were then converted into structured digital wireframes using Inkscape. All screens were drawn using Inkscape's shape and text tools, grouped for organisation, annotated with labels, linked with navigation arrows, and exported as PNG images for documentation.

Conclusion

This experiment demonstrated the complete wireframing workflow using Inkscape as a vector-based design tool. Starting from low-fidelity paper sketches allowed rapid exploration of layout ideas without any technical overhead. Inkscape proved to be a capable tool for converting those sketches into well-structured digital wireframes with precise alignment, grouping, and annotation. The PurpleBank Banking Application wireframes effectively communicated the interface structure and navigation flow, validating the design before any development effort begins. This approach reduces usability issues early in the design process, saving time and cost in later development stages.